

ActInf Livestream #030.1 ~ "How to count biological minds: symbiosis, the

Livestream | 1:51:27

hello and welcome everybody welcome to
the active inference lab today it's
october 1st

5th sorry greeting shannon and 2021

we're going to be discussing this

awesome paper

how to count biological minds symbiosis

the free energy principle and reciprocal

multi-scale integration

welcome to the active inference lab

everyone we're a participatory online

lab that is communicating learning and

practicing applied active inference you

can find us at the links on this slide

this is recorded in an archived live

stream so please provide feedback so

that we can improve on our work

all backgrounds and perspectives are

welcome here and we'll be following good

video etiquette for live streams like

muting if there's noise in the

background and raising our hands so we

can all speak

at this short link you'll find the

calendar of

past and future live streams the first

tab is the regular tuesday live stream

with two weeks per paper with the dot

one and the dot two but then we have a

few other

series of streams related to different

topics so go to that link to check out

what's upcoming

and see if you want to participate in

one of these streams

today in
number 30.1 the goal is to learn and
discuss
this awesome paper on counting
biological minds and we're really
honored and appreciative that matt sims
is here to discuss it with us
so we're just going to introduce
ourselves and walk through any of the
figures any of the formalisms we also
have a few other ideas and questions
that lab participants have raised
so this should be a great discussion
thanks everyone for joining we'll just
go around and
say hi and introduce ourselves either
just an introduction or feel free to say
something that you are excited about
with reading the paper and then we'll uh
all non-authors will go and then we'll
let matt just provide a little context
here so i'm daniel i'm a
postdoc researcher in california
and i think
i'm looking forward to drawing out some
of these threads about collective
behavior and the timeless questions
about collective behavior with some of
the new approaches that are discussed in
this paper
so i'll pass it to shannon
hi there nice to see everyone again um
i'm shannon i'm a phd candidate at the
university of california in merced
um and i've blurred my background but
there is a dog rolling around so
hopefully it's not too distracting
um i will pass it to stephen

hello i'm stephen sillett i'm based in
toronto um i'm doing a practice-based
phd through the uk and
around sort of psychological
perspectives on practices particularly
around social topographies and
spatial meaning making in participatory
theater and the arts
and i'm really interested in this paper
i like the
folk psychology questions that it brings
up and i like this idea of a folk
biology which i found really interesting
and the way that
we have assumptions around that
it sort of ties in maybe some of the
folk quantum that came up in some of our
previous sessions about what people
think that is so i i'm really excited to
see this folk idea being extended
because i think that could be really
useful so thanks

thanks stephen and to dave yeah
yeah dave douglas i'm retired from
programming machine translation
software um now i get to get really busy
working on a history of cybernetic
learning theory project for a professor
at carnegie mellon and all this good
stuff here

thanks steve so

matt thanks again for joining we'd be
really happy to hear some context and
we'll just take it from there and anyone
can raise their hand or type in the live
chat if they have a question
fantastic thanks so much for having me
uh so uh the context in this paper i

suppose it's like i wrote this paper uh
and
during my phd over at the university of
edinburgh i was
supervised by dave ward who's an auto
product and activist uh andy clark
uh who i'm sure all of you know and uh
julian kaverstein who's up over at the
university of amsterdam
and the the focus of this paper
uh was
pretty much my obsession on the notion
of collective intelligence and what
collective intelligence might be
and how cognition
uh has has developed say the evolution
of cognition with with a background of
um or taking into taking into account
the fact that cognition is uh in its
most basic form something that's going
to be collective in the first place
uh so it does pull at there is this kind
of immediate um
tension with how we look at cognizers
and how we look at cognition
uh
in the kind of
folk field with typical
uh say typical cognitive scientific kind
of research i suppose a traditional
should i say
uh and the point of this paper was to
okay but look at
uh the kind of evolution of cognition if
you if you will through a particular
lens using some of the kind of the tools
conceptual tools that were provided by
our seth morty and and uh john men and

smith

and see exactly how this can be

uh formalized and looked at through the

perspectives of the free nature

principle with the process theory active

inference

so that that's that's pretty much it

very

concise summary major transitions which

is the zothemary and maynard smith line

big

qualitative changes over evolutionary

history and then just like you kind of

brought

to at the end how do we land that plane

with maybe quantitative differences or

structural differences in a micro

process theory

maybe active inference under the free

energy principle

um

so shannon or dave or steven feel free

to

raise your hand um

otherwise maybe

a first question that came to mind for

me was like how do we research

or understand these folk beliefs

scientists are folk too

and is this like sort of the mythical

average citizen

what are these beliefs and how do they

come to be that way

why are they the way they are these folk

beliefs about biology because that sets

up a lot of the discussion and what's

relevant of the contributions of the

paper

absolutely that's a very very good
question very good question
uh
book beliefs we have kind of obviously
kind of folks psychological beliefs that
take the form of
actual uh for psychological notions that
take the form of say beliefs and desires
and these things i think uh
give us a good a good
very good manner to in which to actually
approach the notion folks thought
biology also with beliefs and desires
being particular type of constructs
which play an explanatory role
a psychological explanatory role these
are successful in for example explaining
or predicting behavior to us on a
certain level
and i think uh on a kind of first blush
level so to speak and i think folks
psychology and folk biology are kind of
um
bed folk in the sense that
biology is going to be based upon what
we initially
see
uh
via
like in our everyday experience of the
world for example is what we initially
see
so we're going to base the notion of an
individual
for example on the on the ideas of
things that are going to be easily
individuated
uh and like

biological systems that are going to be specifically reproducers and this is where you come into the the notion of kind of the kind of darwinian individual as it was called by peter godfrey smith uh where individuals are going to be demarcated by their ability to be categorized under under three different types of that combined or joint three different three different kind of categories which are going to be they have to involve variation uh heritability and some type of differential fitness which is just natural selection right and uh and this type of individual has been kind of uh saying we've invested into this notion of individual and by the biological sciences uh up into a certain point up into a certain point because it helps us answer helps us approach certain questions and make certain i suppose make certain inferences about about systems over the long run but it falls short and i think this is again one of the grounding points of the paper it falls short and explaining certain types of associations uh and these associations namely are going to be symbiotic associations and this is

a quite important uh i say transition as
as as we spoke about before
uh for the simple fact that without
symbiotic associations we don't really
have a manner i think i think of
explaining

uh

multicellularity

right the the transition from single
cellularity to multicellularity now that
this we've had various theories and i
think one of the theories that kind of
one of theorists that started this was
lynn margulis over at the university of
madison wisconsin

and her research was fundamental
and and when she suggested that for
example you have two

uh

two prokaryotes one envelope the other
through some type of phagocytosis
and they create some type of
relationship over time there's just
thought as ludicrous

and it did really push against this
notion of darwin darwinian individual
uh however over time we've come to
accept something like this uh and uh i
think this is like

the idea that you have some type of
bacteria that was enveloped by a
an archaea

and

in order to explain in order to actually
think about what we have in terms of
this particular type of unit

uh we really do need the notion of

uh moving beyond moving past our winning

individuals and two ways of doing that
obviously i think that the paper tries
to point out this

uh and it's these are not the only two
ways but one way is one way of doing
that is through kind of the metabolic
individual which is something that peter
gabriel smith himself kind of um

uh tips his hat off to

uh which is just to say that metabolic
individuals are individuals which
uh

they can be delineated by their
metabolic machinery and having uh kind
of autonomous metabolic machinery and
other other individuals can play a part
of that machinery

and thus you have uh two systems that
may kind of be
metabolic coupled to one another making
one system

and that's one way to go about it the
other way to go about it is i was
introduced by thomas

and this is a kind of the
immunological version of
an individual or a physiological
individual

now again these are the only two types
of individuals or two ways to delineate
individuals outside of the kind of more
traditional

folk biological notion of individual but
maybe that's something that we can talk
about perhaps there are other ways that
it might be interesting

great points and so interesting how the
map replaced the territory the map of

the sort of darwinian triad
defining an individual in the darwinian
sense and then people project that back
out
and say reality is explained by this
partitioning scheme yes
so very cool steven uh with the raised
hand then anyone else in here or in the
live chat go for it stephen thank you
yeah i'll be interested uh i think this
is really good points about these
different ways once you start to have um
these interactions and these
repres
reciproc reciprocities across p you know
then pluralism starts to come into the
picture yes and i i'm wondering how you
see um
this moving between fields of practice
because i wonder if like people in the
biological sciences and maybe those who
work with agents would really
be getting this right and then people
who have a theory about thinking they
know what biology is
maybe physicists
who maybe think they know what
everything is
um then have you know they have a
they have a biology understanding of
biology
just like maybe other people have a
physics understanding of quantum physics
so
i'd be interested in that and this might
even bring up some work of chris fields
that we had on here as well in terms of
other

ways of approaching this
multicellularity but i'd be interested
in your thoughts first about you know
that
tension even within the sciences
that's a really very good very good
point i do think this is a uh
before i hit that point directly
i do think this is a call for and just
interdisciplinary work
it's an absolute call for
interdisciplinary work this is something
uh that's that's it's
i think
trying to actually
use
psychological notions of biological
notions or physical physics notions uh
we will turn we will have some success
doing these things however
once we run up and there are going to be
certain constraints
uh with doing that and we'll run up to
problems run into problems and these
problems i think can only be
uh
can only be avoided or actually uh
addressed
when
using various disciplines together and i
think this is one of the things that
kind of the free energy principle
actually offers
uh as a kind of it's an opportunity
to work interdisciplinarily
on on various uh various phenomena
various processes
uh and

you're absolutely right in saying that
like okay well
uh how how can for example
in biology you might have
a typical assumption about what
cognition is
so we'll have a lot of biologists for
example there's a
there's a
there's a type of i that we were
speaking about before the before this
there was a
a notion of basal cognition that has
been kind of pushed
uh within the last 15 years and a lot of
various kind of research programs
scientific research programs whether
it's going to be plant biology or
microbiology has
has kind of
fed into this which is it's a lovely
program in the sense that it's
questioning
uh how we actually look at cognition
and these biologists that are kind of
taking these cognitive terms and using
these cognitive terms one of the things
that you know uh
i think that pushes against this is
going to be kind of a traditional
psychology
that uh
refuses to
uh with lack of lack of better words i
suppose which uses to actually engage in
some type of interdisciplinary
discussion
so it's it's a happiest psychology

that's happy using
for example um more focus focus
traditional and folk uh psychological
notions
and
uh with these kind of limitations i
think
we're going to actually be stuck we're
really going to be stuck
um and so
i i guess to summarize the point
is is that
it is a call for interdisciplinarity it
is a call for kind of questioning uh or
putting aside
the fact that we can give
uh satisfactory explanations but whether
or not our satisfactory explanations are
are actually the explanations or if
there is just one explanation uh
it that's that's the question that's the
question and i think that the whole
notion
of uh
the whole the whole idea about uh
biological individuality is that you can
be a
once you take into account various
various angles from various disciplines
it's fine to be a pluralist
it's fine to be a pluralist
thanks i i was thinking of just a
pluralism of what the word folk even
means some people might think oh folk
chemistry that's being wrong that's the
middle school level of understanding
chemistry and then the experts they know
about chemistry that's folk versus

expert but also

it seems like focus being used in a way

appeal to first principles like

appealing to

directly intuitive observable processes

and um what is active inference if not a

first principles approach

so

there's a way to bring in some of the

technical expertise the computational

backbone

but also

fully retain

the conversational and the casual

approaches that appeal and spread to

people

in their understandings of different

sciences

yes so

dave with a raised hand and then anyone

else with a then yes go for it dave yeah

there has been so much good work done by

people outside their specialties i guess

erwin schrodinger may be the most

well known of uh in that

um

but um ben malka the swedish uh

neurophysiologist

uh wrote a lovely little paper from

about 1999 i don't think it was ever

published

about

how the development of the mobile eyes

in

one extremity of the early vertebrates

has forced

neurological developments

that have resulted in consciousness

simply because having mobile eyes
creates such an enormous burden
on
sensation
that you can't just use a simple nervous
system otherwise it's just overwhelming
have you looked at any of mercury
america's uh work
no okay i'm gonna have to send you that
and uh
please do
yeah yeah and um also mark solems uh
who's a direct collaborator with uh
professor fristen just wrote a popular
account
of um
active inference from the viewpoint of
his own work with america and yak poncep
and antonio dimazio on
the effective
uh foundations of consciousness and
especially his own
thesis that
in freudian terms
it is the id that's conscious and the
ego
the cortical and the neocortical
processes
only
gain to become conscious when it really
has to because it's so expensive
metabolically and in terms of
distraction
um
and he's one of the few people who's
actually aggressive enough in trying to
go out and market his
dissidence

from the received

wisdom

and he's also he's also did a uh an

amazing update of the original

um

project for scientific psychology

in terms of fristonian

active inference so

first word the last word of that 1895

freud essay is in 2017

terminology

but still in exactly freud's order of

presentation

wow

yes please do send me that paper i would

love to read it

awesome so

stephen with a raised hand then shannon

or blue welcome and anyone else in the

live chat of course

yeah just uh also

this this idea of the folk psychology

and i agree with your point about

interdisciplinarity and it's i think

it's really interesting is how the

experts perceptions of the world then

gets translated into a folk publication

which then becomes the kind of landscape

which pretty much everyone then

starts to

incorporate and one one that i found

really interesting with active

inferences i was looking back at pre uh

older

ideas of vision

before they discovered the retina and

the image on the retina

and they actually thought of the eye as

projecting out and sort of feeling the
world if you go back like 500 years
and then

we were told by the experts how it is
with the retina and the image projected
like a camera so then folk psychology
moved to that

or folk perceptions move to that because
the experts said that but actually it
turns out that the original folk
psychology or the original perception is
actually more accurate now that we're
looking at circades and

uh all of that so this is an interesting
what is folk and what's expert and where
does it come from
absolutely

it's a it's a both and because it's not
like the retina emits the photons that
it then senses back it's not doing echo
location

but there is

a tale of two densities there's a
generative model and a recognition model
and there's a process that's a
conversation which we're going to get to
with

users and resources distinction so it's
like there is that bi-directional
conversation and then if this direction
is folk and then that direction is
expert

in certain social contexts which
direction is going to be prioritized as
an explanation and the other direction
is going to be seen as a counter current
literally going against the stream
mainstream

so

how does pluralism allow us to recognize

both directions in that case yes stephen

yeah that's a good point daniel um that

that i do a bi-directional and one

question i i wonder sometimes

i often work i'm i'm sort of connected

to people who are i'm seen in the world

of coaching and psychology as well in a

ways

and the arts where these things like

bayesian brains sort of did come through

there about 10 years ago but in a very

but it's it's normally come from

research which has been interpreted into

a book by someone from that field which

then gets pic which is still quite

technical which interpreted by someone

into something else and

by the time it's rolled down

into some other areas

um

these notions can be quite

quite watered down you know um so yeah

this

this question about how to

you know have that conversation

when some of the stuff is quite

cutting edge from even people in the the

fields themselves

um and how to get it used in the real

world is is a big one

absolutely very true so dave with a

raised hand and then anyone else

i think it's in the panda's thumb where

um

there's a rather exasperated essay about

how you'll have an innovator and if he's

really lucky he'll have a few people
that work with him that fill in the gap
but typically and
typically they don't he doesn't even
have one level of competence it just
immediately goes into the full press and
becomes just a complete mush with all
the
almost all the original valid insights
just
buried under
full foolishness
possible so i guess just we paused on
this
slide because this concept of
individuality is a central one here and
we mentioned a few of these different
uh
ways of looking at individuality and
kind of contrasted pluralism the idea
that multiple perspectives can co-exist
and that that's okay versus i guess
absolutism or modernism that one of
these or some other unlisted criterion
of individuality is like the the broader
category or the right one um what is on
the table with definitions of
individuality like why
would it matter in a technical capacity
or for just all people what what is
being discussed or what matters
depending on how we approach this
question of individuality to matt first
and then to
right i think there are a few things
that we can say why it matters one thing
is that we ha
before we engage in some type of

investigation about a particular type of
phenomenal process
it's quite good to know
how to individuate
what those process what's what are
involved in those processes or what that
process belongs to
uh
the relations between elements in a
network pretty much
right in order to understand
those elements or to locate those
elements we have to have some type of
criterion in order to in order to
actually even think about uh the kind of
larger structure of a network and the
relationships between them and i think
this is biological individuals
uh they they actually play this role
and i'm reminded by
something that uh gregory bateson said
right
that in order to actually understand
what the mind is
we have to actually understand how to
delineate things with minds
but those delineations are going to be
completely dependent upon
uh the the processes that we're studying
right
those delineations are going to actually
involve the processes that we're like so
it's going to be a process dependent or
a phenomenal dependent individuation
and that doesn't say that doesn't mean
that uh
these kind of individuals are going to
actually uh

they might contrast or conflict with one another in any type of logical matter but it's just putting in place as a type of pragmatism in terms of uh where we delineate things and i think biological individuals provide us with a manner of talking about for example specific phenomena now if we're talking about replication we're talking about for example competition uh with respect to something like replication

then we might want to actually focus on darwinian individuals however if we're talking about the evolution of cognition or we're talking about the evolution of multicellularity or these major transitions in evolution we might want to think about individuals in a different way

so that's that's that's what i think thank you matt blue and then shannon so hi matt it's nice to meet you i love this paper found it super um inspiring and um you know i think that the concept of an individual um is closely related to maybe the concept of autonomy and also the concept of emergence and so as and especially i mean it's also of course related to the concept of a cognizer but but cognizer aside the so i've heard a lot of definitions thrown around and an individual

um i liked in the paper how you described emergence with the circular causation because that like echoes a lot of um i mean it's the the information theory of internet individuality paper by um krakauer and that group and it just echoes a lot of i mean what's required for an individual so i i just i'm thinking about um autonomy and and it's one of the things that um i mean it depends how you define it and in like the artificial intelligence world there's no such thing as an autonomous system i mean i mean really we program in whether we want it to learn whatever it wants to do but i mean they just don't little baby robots aren't just set free with no goal there's always a self-directed or a human directed goal kind of programmed into this system or maybe i'm wrong maybe there's examples where it's we just create robots for for fun or or artificial intelligence just to create it but like an artificial intelligence system doesn't really have that same autonomy or at least i don't think about it and so i just i'm kind of curious as to you know in the paper you kind of hint it out or i don't know if you said explicitly i was kind of flipping through looking for a quote but you know you hinted at like maybe a cell doesn't also doesn't have that same

kind of autonomy

and

like so i i um

i i want to challenge that

so

in in terms of active inference and the

fvp like i think about it always as a

skill-free architecture and a continuum

and so

why is a cell not autonomous but an

organism is or maybe an organ is is an

organ autonomous or where do you draw

the line and why are you making a line

in the sand around

a system like that

fantastic question and uh i agree with

you i agree with you that's one of the

things in the paper that i think that i

pushed i did draw a line i didn't try to

draw a line that was too

deep in the sand

uh and i think in one of the footnotes i

do say something like okay well i'm not

suggesting that a mitochondria

is going to be something that's

non-autonomous

uh

however

autonomy is something that also varies

in degree

that i think a certain level there's a

certain level of autonomy

uh that it's going to be

uh

providers that provide a a different

degree of of i say degrees of freedom

with which

some system can interact with its

environment in ways
uh and that integra that environment
provides perturbations
uh enough perturbations enough kind of
uh
enough surprise
such that that
that whatever that system is
has various strategies
uh it has to have various strategies
behavioral strategies or we can talk
about this in terms of active inference
uh and in order to actually
adapt to that particular environment now
i'm not this this gets us into this
notion of adaptive active inference
versus
uh kind of like
the non-adaptive active inference
and i think the way that michael and
julian have thought about if i'm not
mistaken i have thought about uh
adaptive active environments is that it
is based on a certain level a certain
degree of autonomy and with respect to
cells
you know
i agree i do agree with you
cells do have a certain level of
autonomy however
cells are again
dependent upon other cells
uh in a way that
other systems other more autonomous
systems aren't dependent now you'll say
of course we're dependent upon our
environments we have to have oxygen
there are certain constraints in our

environments that we have to um
we really have to kind of like our
environments have to be a certain way so
we are dependent upon our environments
but it's it's the distinction i think
between
being dependent upon other systems
uh and cells for example bodily cells
are dependent upon other cells uh not
only uh say that plasma but other cells
in order to actually maintain themselves
but i think there's a difference in
degree so that's that's that's and i
shouldn't again i should impress upon
that this idea that i'm not saying that
cells are not uh autonomous
but most certainly i think there's a
there's a difference in degree and that
difference in degree does uh does matter
uh cells might eat cells might even be
cognitive to a certain degree and this
is something that some someone like mike
11 michael levin suggests and i and i
agree with him
uh
but i suppose the notion about
adaptivity uh
there's a degree of addictivity that
comes along with us uh the emergence of
a certain kind of uh degree amount of
degrees of freedom
so cells are dependent on other cells
but aren't humans dependent on other
humans i mean we're born as an infant
and completely reliant upon our parents
to provide everything for us for many
many many years and and even as adults i
mean still we don't exist in a bubble i

mean perhaps we can go off to a cave and
exist in isolation but but i think
humans are dependent on other humans as
well
so maybe it's a degree of dependence or
not or
i mean i i really feel like there's a
continuum here even in terms of adaptive
active inference and mere active
inference i mean and i fall in with mike
levin in
in a lot of the ways that i think so
yeah yeah i i think i think that's right
i do think that's right i think there's
a distinction to be made also though
between kind of uh
developmental dependence
uh and homeostatic dependence
right and
and i think homeosta developmental
dependence can have an effect on
homeostasis homeostasis
most certainly
but those are two things that
might not actually uh always always
overlap and uh for example sure
we as humans absolutely we have a long
developmental period in which we're
absolutely dependent upon our parents or
being reared by some other human or some
other animal
absolutely and that happens that when we
actually
uh when we when we get older we probably
are dependent also i absolutely agree
with that
but there isn't any kind of development
with a cell

that can be mapped on to
uh dependence or non-dependence it's
always dependent
thanks matt shannon
that actually uh flows really well into
my question blue that um i know your
paper you specifically say we might want
to talk about humans or groups of humans
but we're not going to in this paper
um but i'm going to ask about humans
and so when you're talking about um
this
this
squid and it's bioluminescence um
and they form a temporary unit in which
they are
together
an individual
um and then it sheds the
the the bioluminescent cells during the
daytime and then
they breed again
afterwards i'm wondering if there is a
time like a critical time period at
which
units would have to be
uh have
this symbiotic relationship with each
other or if there's a necessary strength
of the coupling or a repetition of how
many times
um the organisms need to meet up and if
that would change for for this
this animal compared to systems of
humans because there's a certain way in
which um the the humans that you
interact with in everyday life or if
you're co-living with someone even your

metabolic states become co-regulated by
your sleeping patterns with each other
or
stress and hormones between different
humans are affecting each other when
you're together
at a fairly regular
rate and so i wonder
if we might define individuality or
collective slightly differently
depending on the system we're looking at
and the type of coupling that's
occurring
i think that's a great question chad
it's absolutely i would i would say yes
i think this has to be some um obviously
that you know about the the empirical uh
literature in terms of kind of
entrainment
uh
and i think this is like that's a really
good point to make in terms of
having those types of interactions with
humans whether or not
for the for the bobtail and uh vibrio
uh there's a chance that um
like out after the vibra actually
colonized the light organ they're
typically not taken in again
uh that once they're expelled they're
out and they might be taken in by
another squid or juvenile squid but
they're but the vibrio that continued to
colonize are the ones that are left over
uh and
it might be the case that the ones that
uh continue to colonize for that shorter
period of time

uh

might become

the relationship with the script might
change to a certain degree but i don't
know any

uh any empirical evidence and i'm
ignorant about this point that suggests
that's the case but i think that's a
really good really good question to ask
a very good question to ask

so blue and then i'll make a question
i think dave was first oh yeah go for it
dave at them

yeah they're from

dr alan shore has done some um

i think controversial but really

evocative work on

the formation of the not just the
personality but actually the neural
functioning of the newborn in the dyadic
relation the intimate dialogue with the
mother

face play and the rhythm of moving in
and moving back and
regulating each of them regulating each
other's expression of interest and each
other's degree of involvement
and at a really different

scale

um

gerald edelman's

topo biology talks about the cell and
the tissue as being just so dependent he
points out look we have almost no
chromosomes we have almost no genes
compared to how complex we are where
does the regulation come from it comes
from adjacent cells adjacent tissues

right thanks steve blue
so um shannon kind of what she was
saying about um becoming co-regulated
say if you live in an environment with
someone like it's a super nice
transition to something that i have
really been wanting to discuss uh in the
context of this paper which is like okay
you become co-regulated with someone
that you're roommates with and and
co-regulated perhaps by the environment
but also perhaps by the microbes that
live in our environment right so there's
like the entire micro microbiome concept
that is is fundamental and key to humans
and i don't know i mean i'm not like a a
huge studier of symbiosis but
i'm not sure if the microbiome like i
know we depend on a healthy microbiome i
mean for healthy like brain function and
you know just regulation of our own
internal like uh metabolism and so forth
and so
is that a symbiosis
or not um also like the concept of the
holobionts which is like introduced by
fields 11 if you've read that paper
they're the scale free biology paper i
mean
it is like to what
and that they they talk about um
physiological individuality there as
well so is that symbiosis is it not is
symbiosis it's also a spectrum or is it
like a you know a discrete category of
things and i also think about like
symbiogenic viruses
which like you know the very early

viruses that it's like you know
effectively like instantiating
speciation right like the viruses insert
into the genome and you know developing
species and so forth and and it's
beneficial to the virus and the host
right so so anyway i just wonder like
what are your thoughts on that or or or
um if you have any insight there
absolutely that's again a wonderful
question boo um with respect to kind of
symbiosis i think there's some there's
been
some work on that and some very i think
uh convincing work done by queller and
strassman
uh
who suggests a particular type of xy
locating symbiosis on uh kind of an axis
pretty much so you have uh cooperation
i believe on one part of the axis i have
to have to look it up again actually i
should look that up um
but it's uh you have varying degrees of
symbiosis
uh and again symbiosis in the in the
particular paper that we're discussing
now i've concentrated on mutualisms but
you have various forms of symbiosis uh
that i think you've actually pointed out
previously uh that can be kind of
modeled
quite well with kind of game theoretic
uh and this is going to be is going to
rain from parasitism
to uh communalism to kind of um kind of
mutualisms
and i think like these are all again

degrees and these degrees can change
uh there have been quite a many theories
i suppose and i think squadron strassman
have also suggested this that uh
depending upon the condition of a system
uh with respect to another system say
their symbiotes a parasite can turn into
a mutual a parasitic parasitic
relationship can turn into a mutualism
and vice versa
so these things move these types of
things move uh and if you think about
this is an automatic thing that we can
even apply this kind of cellular biology
when you think about something like the
the origin of cancer
right
when you have
cancerous cells
uh
they're they've actually grown within a
host
right they're part of a body
uh but end up being some type of
competitor
and i think this is quite interesting to
think about this it's like through the
lens of symbiosis obviously it's it's
not that's that's not a symbiosis but we
can use symbiosis to kind of think about
that
uh
but with respect to kind of
uh thinking about
host cells and parasites our whole cells
and mutualisms
uh there's like
there's there's many many different

kinds of ways

areas of the spectrum that one can

actually analyze uh our relationship

with our own cells and our relationships

with uh with micro microbiota microbiota

like scott gilbert obviously you know

and scott's work is is quite incredible

he's the one i believe who introduced

the notion of the hollow biome

the hubble genome excuse me in halloween

and uh

i think this this notion about uh hollow

buoyant is specifically

uh

it includes symbiosis but includes

actually a kind of community

a particular kind of community of

various systems

uh that themselves qualify as one system

can qualify as one system or having you

know if you would put it in a different

way they they create a stable state

right a steady state themselves

and uh one that's kind of resistant to

perturbation

uh and

that's that's the matter in which i

think it's like it's it's a very easy

matter in which i can think of uh

the

blind as as an individual

uh whether or not one wants to call

two

two symbiotic say symbiotic uh systems

um

that are in a mutualistic relationship a

hollow bond that's another question but

i think typically it's been applied to

um more than two systems
thank you matt
so for
shannon's questions about you know the
co-regulation and humans and trust it
made me think about how
one-off games
lead to kind of community or
population-like game theory whereas
repeated games over evolutionary time
like in the case of the squid and the
vibrio or in the case of the eu social
insects that leads to
organismal-like dynamics and then what
you just described and what um levin
also got at in the paper is the
breakdown
of for example organismality into
population of cell versus population of
cell dynamics so it's absolutely um a
cool framework and then also a really
important part on figure four
is you have the peak of integration and
then greeting serval
the peak of integration
um and then that's the y-axis so it's
like instead of saying we're gonna say
which organisms are tall or short out
there in the world it's like you do the
population level survey and they range
from you know two to four feet and you
say okay it's in the upper 50th
percentile it's tall like we're
renormalizing
our comparison stick to
the maximum integration of the squid
bacteria so we're not comparing it to
some other system we're re-normalizing

and we're able to have a system-specific understanding here it's relatively more integrated here it's relatively less it's a second question whether the peak of integration of the squid and the vibrio is like the integration of some computer program with another computer program maybe it's apples and oranges maybe we'll have the right framework that allows us to transmute apples and oranges but here you can just relativize to the system itself and that's like very specific and powerful so stephen with a raised hand and then anyone else

yeah

i'll give you a chance to get a breath because we've thrown a lot at you but um the you mentioned allostatic and homeostatic forces if i imagine there's the the so the so you've got that difference there between something trying to maintain or trying to try to move forward and change

um

and i'm interested in in how that reflects um

so for instance i suppose you could have within a set of cells you could have some cells or some

organs you know trying to maintain homeostasis and others

say even maybe a parasite is trying to anaesthetically

enter that world and create a niche for itself

um and i think that that that question about

you know

that that balance between the two is a
useful divide or a useful way to think
about the same

cell and of course in cells they change
to become different types of cells so
that's something that

whole organisms have less ability to be
completely

completely different types of organisms
you know they can't um you are what you
are i suppose that's one defining thing
about but you can still change your
focus from homeostasis to allostasis i
think that is quite common and i wonder
what your thoughts are around that um
and how that that itself could be a
useful

way to frame things

that's really interesting

uh the the man in which i'm looking at
kind of a homeostasis uh and uh kind of
coupling of homeostatic regulatory
systems to allocate regulatory systems
is based upon kind of

a more traditional idea that homeostasis
might be considered more reaction or
reactive

where allostasis is anticipatory

right and then allostasis takes a
particular form it's quite interesting

with respect to in the context of active
inference because we talk about expected
free energy minimization

and this is something that uh one can
tie together with the notion valley

stasis

um

where

uh and when you start to think about
kind of homeostatically coupled systems
one system

uh with respect to kind of symbiotic
systems might be playing a kind of a
larger rule uh and being able to make
further and further of being able to
make inferences further and further into
the future and i think this is something
that you've you've suggested you've
looked at previously actually on a slide
uh mike levin's cone slide that blue
and daniel actually had up previously
and i think this is a quite interesting
manner of thinking about

uh kind of a transition
perhaps we've always been allostatic
systems living systems have always been
had certain of certain uh capacity to
anticipate

however when we start putting systems
together

we have a larger capacity or more
capacity to anticipate further and
further in the future and i think one of
the dimensions on this particular cone
is this dimension of alice like a kind
of outstanding control

uh this is this is my interpretation of
michael's work uh and

i think this is that's very important
and again you'll see that there's that
kind of

balancing that if one is
given a type of approximate bayesian
inference that's going to be uh you know
what we suggest is going on uh you have

to actually balance this with priors
which are going to be respective to the
past so you have another type uh another
axis on the cone
and as these things expand sometimes the
past is going to uh be a bit deeper on
the cone than than the than the future
but out of stasis is on on the top there
and i think that like with
with multicellularity
with uh
with kind of
uh the interactions more interactions
perhaps you will see a capacity a larger
capacity to go upwards
and in the kind of future oriented
direction
and this is this is something again
uh and i i appreciate blues bringing
this out this is this is uh something
that's again a matter of degree it's not
a it's not a it's not a line that's cut
it's a matter of degree
right
and so
uh
just simply we know that there's certain
types of
uh microorganisms that that do
anticipate
that do like right uh and
uh this is kind of outward anticipatory
behavior whether it's i think it's going
to be uh
phaser and polycephalum which is uh the
kind of
plasmodium slime mold
uh various experiments in 2011

uh had suggested that you know

uh fuzzerum can actually

anticipate

various periodic uh bouts of dryness

this this type of thing this is an

outward periodic behavior an outward

anticipatory behavior

uh

that's not to say that anticipation is

not happening occurring prior to that in

the internal dynamics of systems that's

just an uh kind of an outward

manner of kind of measuring it

um

again i i do think that this like it's

it's a quite important notion allostasis

uh and

it's perhaps all systems again i i would

be

safe

at least my position now is that all

systems are pretty much allostatic

systems uh that kind of enslave

homeostatic regulatory systems

uh to maneuver

and

invest their

metabolic machinery and metabolic

capacities

uh to a particular type of

uh state that perhaps they haven't

reached yet considering

uh you know a manner of of trying to

kind of avoid perturbation

thank you

matt very similar to mere adaptive

inference being more homeostatic more

like the ball just in the bottom of the

bowl and then increasingly anticipatory
rosen's ecology adaptive
active inference stephen go for it
yeah this this ties in actually i'm a
lot of work around um coaching funny
enough the world of coaching and um
team coaching is like and team coaching
so and i think this also brings in you
know these other types of clusters
groups
symbiotic
um relationships beyond maybe what we
might just think of as a system you know
so the idea of a team
um in a as a
in a systemic context
um is really interesting because
they
and this might be how allostasis in a
social context starts to enable that
larger horizon so each
even allostasis is sort of stasis and
then
working together you may start to break
your own need for stasis you know and it
just
becomes some other organization and i
think it'd be really interesting and
looking at how
the way that team coaching either for
performance because there's performance
orientated team coaching or there's
developmental orientated team coaching
but that idea of and then the enabling
context
becomes what we would call the
ecological niche i suppose
becomes

um

as much the focus of attention

as the system in that makes if that

makes sense and i think this this might

tie into what you're saying

thanks steven so dave i know that you

had a little bit to share but i'll just

bring up one

point related to this definitions of

individuality and then several anything

you want to ask feel free to raise your

hand

we hear a lot of terms like autonomous

and self-organizing which sort of

highlight the um

ability to cleanly partition a system

like people will say the embryo it's a

classic self-organizing system

right except for the context of the womb

and so there's sort of on one hand the

desire to individuate and partition and

look at agentic systems agent-based

modeling autonomous and self-organizing

and then that is critiqued by

the ecological perspective or the

relational perspective so there's sort

of a dialectic between like it's all

individuals all the way down and it's

all relationships all the way down we

see that debate in particle physics

we understand it less than we might some

of the biological systems are in a

different way but it's um like two

pillars that we're navigating between

and it is a continuum

in between so that's why it's such a

nice discussion

so dave if you're around

um feel free to um any time you're ready
you can go for the conversation theory
otherwise blue go for it
okay sure um now slide yeah slides 18-19
we inserted that into the slide show
that we prepared for dr sims
presentation
and these slides bring in some matters
that he doesn't address in his paper so
to justify that but not to paraphrase
the insertions
the free energy principle and active
inference jointly constitute a
relatively mature macro level framework
for describing living psychological and
other complex self-stabilizing systems
as of the moment no comparably mature
micro level framework has been both
proposed for and coordinated with this
macro level of analysis
i suggest that the basis for such a
needed micro level framework was
proposed by my teacher the late
cybernetics
professor gordon pasque of brunel
university in the form of conversation
theory
matt's essay how to count biological
minds considers the puzzle of whether an
individual can come into existence and
pass out of in existence repeatedly
that's account of this puzzle and please
excuse me if i stray from over
simplification into parity his solution
in terms of degrees of integration
is a perfectly acceptable insightful
account
at a macro or molar

level

the metaphors of quantity and elsewhere
probability nicely express falsifiable
common sense attributes of real-world
processes

but if one states this paradox the
paradox of emergent but transient
individuals in terms of conversation
theory then one speaks of a conversation
recurring over time

between the same conversants
this terminology makes the proposal more
intuitive less shocking and more readily
assimilable for readers

better yet familiar ideas attention
rhythm of discourse punctuation of a
conversation through understandings or
through agreements to disagree they all
play their roles

now what place would be occupied by the
special disciplines that have enjoyed
detailed treatment by conversation
theory as a micro theory

during his career professor pask carried
out an array of experiments studying
human learning of complex subject
matters configurations using
configurations of software and hardware
to administer and collect data from
those experiments and develop a
vocabulary allowing elegant description
and restatement of the results of those
experiments

moreover he developed a theory of micro
level learning teaching

as a single literally irreducible
complex of activities

regardless of whether they were realized

by living organisms by machinery or by

complexes of those two

fabrics

i see many deep and broad analogies

between conversation theory and active

inference

each addresses important aspects of real

world phenomena that assist in a

systematic way

such that each treats of matters that

are not so well handled by the other

i believe that active inference and

conversation theory can enrich and

streamline each other

by carving natural science scholarship

closer to its joints

very nice

thank you dave so maybe matt any

thoughts on that and then we'll go to

blue

i have to say i have to say that i'm

absolutely

uh the

the cybernetic theory that i've

concentrated upon have been most

influenced by

uh it's it's very limited

and conversational theory it's i'm

absolutely interested in seeing

uh

more of this more of this

i suppose analysis and understanding

more of this analysis so i'm quite

ignorant about conversation theory and

it's like he most certainly piqued my

interest

um

yeah let me let me

think about this for a few minutes i'm
still kind of internalizing what you've
just said

yep and one note on the conversation a
lot of times cybernetics falls into that
individualism mode like it's the one
self-guided missile it's not the
conversation between the flock of birds
so cybernetics um isn't just about
individuals it can be about collectives
and it was our colleague alex who told
us um via one of his teachers levinchuk
just yesterday that um cybernetics died
when it could explain homeostasis but it
couldn't go beyond

so we're talking about the transition
from individuals to collectives
and from mere homeostasis to adaptive
allostasis over developmental and
evolutionary time scales and so it's
sort of like you can have homeostatic
individuals

and then they can have emergence
towards allostatic collectives
and then what is that
process of emergence
it's a conversation

or potentially something like that using
our folk understanding
of interacting pieces which are us
blue

so um my question is maybe technical but
also uh maybe serval has some insight to
provide as well um so i wanted to ask
here in the beginning of your paper you
talk about
free energy being an additive quantity
so

the free energy that arises at the macro scale is only the free energy that there is for the ensemble and may be quantified by summing the free energy associated with each constituent micro scale markov blanketed system so is that true and and like does that account for emergence because i feel like there has to be some like greater free energy minimization and so like i i'm not i'm not well versed in the technical like mathematics of the fep um because i think about you know like just in terms of the bioluminescence of vibrio fisheries right right there's like a criticality that happens like almost like a phase transition that the system undergoes to become bioluminescent like after there's a certain concentration of bacteria in the light organ and so like does this criticality this phase transition does this not like to minimize free energy to a greater degree like or is it really just the sum of its parts wonderful question really wonderful question uh i do think that okay there's a might be a distinction in terms here in a distinction in terms of like what uh the free energy uh of the actual constituents as a group have as an ensemble have

uh

and a distinction between that for

example and

bioluminescence as a particular type of

free energy minimizing strategy

uh

the

it's a i think it's a fantastic question

i'm getting the the the notion of kind

of uh

you know the summation of free energy

from uh palacios

and aronstead

uh and

this i think that the pretty much the

idea is that

i'm not sure if there's going to be any

well i'd like that maybe i can ask you

blue do you use do you see that there is

some type of inconsistency between

saying that

uh

for example

uh

free energy is going to be an additive

quantity

and

and the fact that there can be some type

of emergent property that is not going

to be reducible to

uh

say

the the constituent relations and and

the elements

do you think that there's some type of

uh inconsistency there

well so i definitely do think that free

energy across scales is additive like i

i agree with that wholeheartedly but but
i don't think that it reduces to the
mere sum of the parts
and i mean this also um
it's the the work the recent work of
rosas and also eric hole looking at the
partial information decomposition over
the um five metric that's used in
integrated information theory of
consciousness and so this kind of speaks
to that there there's
it has to be some mathematical i mean i
just feel like i don't know that there
has to be but i feel like there should
be some mathematical
um
quantification of more so it should be a
greater degree of free energy reduction
when the system is more than a sum of
its parts and so i think that that's an
argument that you make in the paper as
well that it can't be it's it's not
reducible to just the component parts so
is the free energy then just reducible
to the component parts here because that
to me seems inconsistent
okay
okay the matter would i think uh
i i see absolutely see your point no
matter what i'm thinking about this
though i said okay well
uh
there's a very informative paper by
santos he's a philosopher he he actually
deals with immersions uh and
uh and
it's a really nice way i suppose i
really like this uh the paper that you

suggested about kind of using the five
metric

uh and various manners in which to like
to do like decom information
decomposition

uh

over various time steps

uh in order to

think about

some way to some some metric for
emergence

i think there's been various metrics for
emergence i haven't picked my favorite
metric that's something that i'm still
getting into

uh

but i think this is it's quite

interesting to think about okay well

uh what makes an emergent system is an
emergency system going to be more than
it's

the kind of relationships and elements
of a network

and what makes it so what makes it so
it's going to be the relationships
themselves or

or is it going to be something uh the
combination of the relationships or is
it going to be the interaction between
the system the emergent system and the
environment and i think when you start
thinking about the interactions between
the emergency system and the environment

uh

there's there's a distinction between
what free energy is going to be
minimized

uh or

and the free energy of the system itself
does that make sense

is it there's a distinction being made
uh and perhaps i'm wrong about this
but

emergence is not something and one of
the reasons why you have an emergency
system

it's going to be for the simple fact
okay well you have
uh

it's not only the interactions of the
parts and the parts or the elements it's
going to be it's going to be the
interaction

of a particular type of persistent
and i think this is the this is the idea
you have a persistence

uh with a particular environment or with
an environment and it's the
perturbations within that environment
uh that occur along with that system
in which kind of free energy is going to
be a free energy that might have to be
minimized is going to arise and i think
it's like so

um
you might say of course that the the the
system itself

uh when it has component parts and
relationships between those component
parts is going to be uh that system can
be thought about in terms of
an additive or summation of the free
energy of the various parts and
relations in that system but that's
distinct from that system's engagement
or interactions with this environment

then you'll have extra free energy
you have an additional amount of free
energy and maybe that's what that's
that's that's another way a matter of
thinking about okay well uh and an
emergence
as measured i don't know if you can
measure emergence in terms of free
energy but of course we're going to be
able to measure it in terms of some type
of mutual information
very nice thank you that's i think that
was
a question there i don't know if you can
measure emergence in terms of free
energy like you said and you might have
to look at something like mutual
information and i was thinking that
perhaps
the free energy that's minimized by each
individual system
then when they become a symbiotic unit
or something might end up being
identical
and again
i'm thinking of humans again but if
humans are walking and they happen to
start walking in step
um that's gonna look like an identical
oscillation
as as they're coupling with each other
so there's might not be more free energy
that's minimized by this coupled system
walking in step together
um but it might look like this the same
free energy once they're a coupled
system
and one thing that we talked about i

think in the dot zero or around there is
there's situations where the cooperation
the coordination increases synchrony so
just increases the mutual information of
the humans walking in step but also you
could imagine
anti-phase relationships which is kind
of like division of labor or task
allocation where the emergence is not
due to um
copy and conform but the emergence is
actually you know you go left i go right
and so in both of those cases
there would be a change
in the non-equilibrium steady state
which is what is introduced throughout
the paper and so that starts to build
the road towards quantitative
discussions
and you could take a system that
you suspect already has emergence in
nature
and remove a part that's kind of like
the gene knockout approach so gains of
functions and loss of functions and all
the other ways that we study
function
can now have something like a common
grounding potentially using these
metrics here
so
dave and then
maybe matt if there's a response
otherwise stephen thank you
okay now blue asks
is free energy additive
there's an exercise that i'd suggest
it should bring that question to a more

down-to-earth and more

poignant focus

take all the discussions you've heard of

free energy and substitute for the

phrase free energy the word

anxiety

it already is that way dave what do you

want me to do

[Laughter]

what would that show though or do you

think it's a natural fit or could it

could be could we put any word there

what do you think it means

well so just using anxiety like it

totally leads to the mob mentality and

like ultra paranoia and conspiracy

theory so like definitely additive

yeah

possibly multiplicative though like

could be exponential

[Music]

yes steve

yeah there actually have been uh

clinicians that have

really used free energy although they

didn't use the term as probably before

the usage was was coined

um

william glasser in control theory and

his earlier work uh really has this

quantum of anxiety

that these patients had to manage these

tended to be out of control criminal

adolescence so they were pretty screwed

up

um and he found ways to get them to rein

themselves in and

control themselves but

think about the blitz effect london

blitz

you had objectively tremendous stress on everyone all the time

but there was almost no hospitalization

for shell shock among civilians

including the air raid wardens that were

in digging you know they a building

beside them get bombed and they'd rush

in and try to find people and drag them

out extraordinarily low rates of

psychosomatic disorder

there's a certain there's multiple paths

you know people people's brains are

really complex

there are certain ways that people

actually do get shocked into rationality

i've been in situations where the right

thing had to be done right now with no

hesitation no anxiety no blame

uh

just do the right thing and there was no

question just do the right thing uh

whether it's whether it puts me in

danger or not you know a child's

endangered you do the right thing so

it's not predetermined uh but typically

yes what does happen is

uh a thing called reorganization it's a

technical term that

glasser uses it essentially means going

crazy and in the really worst case the

reorganizations get faster and faster

and faster and more and more drastic as

the personality becomes more and more

simplified and that's something that

kurt levin or lewin worked through in

his work on driving children crazy by

giving them more and more complicated tasks and under more and more anxiety and he was able to make the kids personalities and thinking degenerate and simplify more and more and become more and more infertile and you're right blue that is one of the things that happens in a mob the it's terrifying to be in an even unified angry mob where everybody's mad at the same thing and they're probably not going to attack each other it's terrifying and people that keep going into that situation things happen remember um uh um the presentation two weeks ago on the mutual suppression of the limbic system the reticular activating system on the one hand and the prefrontal cortex on the other you get these things that look like beliefs they sound like beliefs they act like beliefs but they're not beliefs because reason has never been invoked in the biological decision whether to incorporate this magician's spell into my feeling and my actions do you have to have reason to have a belief dave oh yeah in the sense that uh uh philip

uh

gerunds uses it

yeah

yeah you have i mean that's that's kind

of the way he defines it a belief is

something where the prefrontal cortex

has been involved it's been in the loop

and there's way to bypass that um

i asked him whether he's worked on the

social psychology of this or the the ego

psychologist he says no no

i have enough trouble worrying about

people with damaged brains and how that

shows the bypassing

of rationality in the formation of

delusions but i think it works the other

way um you may have seen my my

shotgunning of the apathetic apothemic

brawl

by uh dr

nathan rosenberg or rather gabriel

nathan rosenberg he's an academic and i

think kentucky he says look this is

what's happening you've got this person

with a

a non-existent personality who's doing

what the mob tells him to do through the

mob's megaphone called fox news

why do we like him so much because he's

less than any of us he will always obey

us so we'll give him power and then

he'll give us what we

feel like

uh yeah and there's no reason of

ultimate it's very much a bypass yeah

rita labonz uh

study and mackey's study on

the madness of mobs

and the mass production of delusions and
crazes
and
mccarthyism and all this
unhealthy stuff
thanks dave and that was the guest
stream 10.1 with philip guerins and so
just to highlight one point and then
stephen it's we're really talking about
control theory
maybe the first pass the first term in
the taylor series
is about control i mean come on narrow
the error and that's kind of definition
one
control as homeostasis even if there's a
shifting set point but then
again it's the ecological and the
relational insight
and just the emergence of interactions
that
control takes on this broader
anticipatory
usage to control your body temperature
is to sometimes do
a lot of movements that from that first
pass look like a non-sequitur like
sprint so that you can get into the
air-conditioned room
that's the expected free energy through
time
so now we have a common grounding for
talking about control when it looks
homeostatic as well as control when it
looks allostatic or even innovative
so it's very important to pull out some
of these threads and also recognize
their cybernetic origins and other areas

too

so steven and then anyone else

yeah um these are all really good points

i'm curious when we talk about free

energy minimization

i think it's it's interesting to talk

about that from a systems perspective um

however if we want to get back to the

kind of organism

so the key thing with free energy

minimization

that occurs at different scales you know

physical scales number of

units scale time scale

you know teleological scales we think

about these different scales but

that reduction there has to be a way for

that reduction to be meaningfully

engaged

through actions or interactions like

the metric ultimately what's going to be

filtering through

probably in some noisy maybe backdoor

way

is that um if it's going to be sustained

over time there needs to be some

meaningful engagement over time so for

instance you know the idea of the path

that we walk

starting to wear away the grass in our

niche

well it's because we walk on the path

more than

once that we have an ability to observe

that you know if we only ever walk that

path once it wouldn't

have a way for any variation to be

noticed but we notice the path being

worn
by agents over time so
when we're talking about these different
scales
these variational dynamics in free
energy it's not just a reduction because
we don't
get
a number each time saying this was lower
than last we just get
some
new state in the world and we have to
infer if it's changed from some other
state in the world
and we it may go up it may go down so we
may infer
at times the free energies
has not been minimized and we might
take but overall static processes
for instance if we have a big room
that's not tidy and we have a group of
people and they want to organize it into
a studio space
you know they might work together
initially putting in a lot of extra free
energy time and
and then eventually they know that they
will be able to minimize
their sort of systemic
engagement as a group
while it's not necessarily that they're
each minimizing free energy as a system
in the so this is almost where maybe
challenging to get out of the biological
sense and maybe come into some other
disciplines where
it's not all a system it might be more
of a systemic context

which is then more teleological so i'd
be interested that that question okay
where does this crop through this idea
of
being able to to have a pragmatic metric
that links the non-equilibrium steady
state of the organisms
phenotype or individual so they're
actually able to then
infer something from whatever is
happening to the free energy out there
or in in there or around them
and how that changes things if we if we
frame it away from
systems per se to something more
systemic
different uses of the highly poly semis
word system systems of systems for some
people are systems so that's why we
um increasingly will try to be
exploring the active inference ontology
so that we know which pieces of the
puzzle we're talking about and then that
will unpack so interestingly into
different disciplinary contexts stephen
so nice
points um so in the last half hour
i think there's a few different things
let's maybe get to one of the key
pieces and specific contributions of the
paper which i would love to hear matt
your thoughts on which is just first
what's going on with users and resources
and second what is the difference
between uni-directional multi-scale
integration and reciprocal multi-scale
integration
right um

those two
notions i suppose of unidirectional and
multi-directional
reciprocal
and users and resources they're very
much so connected
the idea of a user we can think about
this and i think stephen's uh
to use stephen's example of an elephant
path i believe that's what these things
are called sometimes so you see these
paths that are treaded
and are typically you know
used to refer to when we're thinking
about kind of nice construction like
past phones construction

[Applause]

[Music]

this is something that is in the
environment that's is being used say is
a resource that's being used and it's
the effect of previous action on the
environment

uh returning to the notion of so that
that that would be a resource

uh

used by a particular cognitive agent or
an agent that's engaging in adaptive
active inference uh

so

number one it has to be recognizable
uh and it has to actually in order for
the system actually to engage with that
particular resource it has to be
recognizable as such

um

and that recognition can come through
for example epistemic foraging uh or

actually just uh be applying particular
kind of a plan a model to a particular
type of uh resource that has been
applied previously
um this is also the case with something
like stitch merging right
where you have uh something
uh that's it's quite lovely where you
have for example ants leaving pheromones
within the environment uh and that
pheromone acts as a particular type of
mediator
uh directing
further behavior by agents of a similar
type
right
and that's again environmentally
mediated communication
or or what they might call an
environment environmentally mediated uh
you know
uh
conversation between between various
elements of a system
uh and
the pheromones themselves of course are
resources
in the environment that's going to be
perhaps uh contrast with the idea that
uh not all resources need themselves are
themselves
they have to be limited to being
resources they can also be users also so
uh to to make a strange case we can
think about uh stitch mercy actually uh
what if it wasn't actually the pheromone
that was being used but the other act
the presence of the other ant that was

being used the interaction with the ant
itself
so here
the two
uh using using using perhaps and perhaps
you have to tell me if i'm using this
wrong that's terminology but you have
two discussions
right a cinder and receiver both are but
both active sensors and receivers and
actually they're communicating something
they're having a conversation uh but
there isn't any mediation in between the
two uh that can be offloaded onto
uh in this particular situation
offloaded into the environment
so the signal itself is going to be the
direct kind of interaction between the
two
uh rather than a pheromone now that in
that case we can map that onto something
that would be a kind of hypothetical
case of reciprocal multi-integration
right uh
and again
uh stigma rg just happens to be
uh typically and in each cases in in the
way that uh people degrease uh suggested
it when he was studying termites
uh this is something that's actually
involving a particular type of signal in
the environment that's offloaded to the
environment so it's going to be
naturally something that's going to be a
unidirectional multi-scale integration
between a system of team very systems
whereas for example
when you have direct interaction between

two cognizers

or two systems

uh two adaptive systems

uh

they can also of course use some type of resource there can be signaling and this is something that of course happens with vibro fisheries

uh

and the bobtail squid but there's also going to be some kind of other direct interaction between two systems and that's uh that's kind of the reciprocal multi-scale integration

awesome summary i believe the technical term would be converse ants

for that kind of a conversation

but it's a great point how um when you have an adaptive active inference agent and then a niche that's being modified stigmatically that does mere active inference like the spider's web

that's unidirectional but still

multi-scale

whereas when you have adaptive and adaptive in discussion in conversation even again if only in a folk sense you have adaptive on either end and it can

allow for a new

non-equilibrium steady state to arise so not that there isn't a non-equilibrium steady state of spider plus web rather than them separate in fact there will be but it still is a different type it's like a double directional or reciprocal stigma g so just such a cool way to connect that

blue and then stephen

so

like again this like line between mirror

active inference and adaptive active

inference like is it like makes me want

to attack it a little bit um i i think

about uh the niche modification that for

example humans do is extensive and

transcends

like generations and i mean work it like

contributes a lot to the collective

intelligence of the human species and so

forth and

i also think about um

in terms of like the gaia hypothesis

right like so if

if the world is one giant organism and

this is something that we talked about

with mike levin when he came to the live

stream so if the world is one giant

organism

is the niche constantly

adapting to the species that are that

are in it like is this a continuous

adaptive active inference and just in in

terms of transcending time scales you

think about like these you know

symbiosis i guess they're symbiosis but

like relationships that develop between

you know like

orchids and their pollinators for

example so i mean these are intense

co-evolution things that happen over

many many many times

time scales and i mean we can think

about the plant as a plant a cognizer i

mean that's debatable

i would say to a degree like i'm not i'm

not not binary here like yes no is not
not really a good answer
but but is the the niche may be
constantly adapting to
systems to a degree
i think that's a great point really
really great point um i do think the
niche is adapting to systems absolutely
but i would say that the niche is
adapting in a different way
and we can we can use
of course we're going to have to be
specifics because this whole
adaptation takes a policy in a sense uh
adaptation in in terms of what the niche
is doing if the niche is not itself
okay perhaps you want to i can be
accused of being biased interested in
one sense if the niche itself isn't a
biological system
uh
then
we can kind of
perhaps push against this idea that it's
adapting in the same way that a
biological system would adapt to and
this
may be one way of phrasing this uh i'm
not sure if this is going to
hopefully make a bit clearer
my my point is that
biological systems one way of
understanding them is that they have set
points right moving set points
throughout a particular trajectory this
particular trajectory might be
a trajectory across face space
throughout their life so you have a

moving set
and this moving set
is something uh that's organized uh
around a particular type of phase-based
organization uh and it's recapitulated
within a particular type of phenotype
uh defines the phenotype it
characterizes a phenotype or a genetic
model
uh with a particular say a
a field of dirt
we might be able to
actually walk across a field of dirt
or a set of cement stairs
the uh it might actually over time for
what a few hundred years you have these
interesting stairs at the castle of
edinburgh that actually dip it and these
dips in this in the cement are evidence
that people have walked there before but
we tend to walk where other people have
walked and this
perhaps one might call that an
adaptation
to a certain degree but i think that
might be pushing the limit on the word
adaptation given that that set of stairs
does not have a set point it doesn't
have uh it doesn't have the same type of
uh it's not the same type i suppose of
of of self-organized system it's not a
self-organizing system in that sense
uh
and
as such i would think that okay well
uh maybe that maybe the niche does adapt
to a certain degree
uh but there's an adaptation with

biological systems that are
self-organizing and have organizational
closure and i think this is quite this
is quite important
uh that is different than
say um abiotic systems and this brings i
suppose we i would love to talk about
this
in a bit is that it's a question that
blue brought up in the last um
in the last kind of uh
live stream
was
the the manner in which we can actually
understand
uh
systems or i identify
particular types of individuals uh that
are involving the environment so it's
it's not actually saying that the
environment is separate from the system
otherwise we wouldn't actually be able
to actually talk about the emergence of
actually living systems we can get about
we can get into that later though
so i have to just push back on you um
so think about the earth and the
atmosphere of the earth okay now this is
not necessarily a biological system this
is nitrogen carbon dioxide water clouds
etc right but here we are humans
releasing tons and tons of carbon into
the atmosphere
totally changing the the the
set point as you described it like the
homeostatic set point or whatever the
non-equilibrium steady-state density
that the atmosphere of the earth

currently resides in and has resided in
for x million years right so but we are
changing that and we are seeing the
earth adapt to
us and the changes that we're making in
it just in terms of hurricanes storms i
mean the weather patterns are completely
and and they're destroying large amounts
of humans right
and drought famine all of the things
that are going to
cascade through climate change as we see
it progress so is this not adaptive
active inference on the part of the
earth or or what do you think of that
uh
really interesting i think in order to
make that particular type of assumption
one would actually have to
you know ascribe to something like a guy
hypothesis
like lovelock sky hypothesis or the same
type of thing that then margulis was
into later in her life
um
and i think that's that's that's an
assumption that has to be made
uh given the fact that okay well just
because uh response happens chemically
doesn't necessarily mean that it's not
actually an adaptive response we
you know uh gene expression is
biochemical adaptation right so it's not
about just being chemical
uh using chemical or or something like
this uh composition changing molecular
structure as some type of uh criterion
for adaptation but there's a i think

there's an assumption that has to be made about what the system is there and if you're thinking about the earth as a system

uh

maybe the earth is adapting to its constituent parts

right

uh but that would involve that would involve

uh intrinsically the earth being a particular type of living system given the fact that it has living parts

one other quick thought on that is as a model comparison framework so go back to spm dcm all of our categorical model comparison techniques in the toolkit we can actually evaluate

should we have cells as the units and then organisms then the each system in the world or should we use a different partitioning which one of these models is explaining predicting designing controlling better

or which one is more parsimonious i prefer simplicity over utility or i have a function that balances simplicity and utility there's all kinds of frameworks that's kind of part of pluralism is starting with and being with

that

multiplicity of answers

and then knowing who we are and what we use to decide on which models to use so

stephen

you know i'm very much like to keep that ecological niche in the picture

so

you know if it's organism in niche and
groups of organisms in niche
and then watershed

a micro environment or micro climate
as a niche

i think that kind of becomes like a
defining organism or

unit in a way is like watch the
watershed what's the kind of
climate as experienced by that
ecosystem

and then of course you've got many
ecosystems around

the world

from the arctic to the antarctic you

know and it will be in between

now they might all be

collectively

affecting the broader

carbon dioxide levels of the broader
ecosystem

but when but how that then jumps to make
the whole world an organism

becomes for me slightly challenging

right because it's it's always it was

always a niche up to then and then

suddenly it jumps to be an organism

because we look at it from space and we

see a nice round ball

and it sort of looks alive so i wonder

whether there's a so one one of the

things i'm really interested in is

is like you're saying how things get

ascribed

to

the niche and we we talked about the

path walking on the path and

in a way like
that dispersive
thing animals can sort of also intuit
some of that as well from the
affordances and the directions
and again bringing stuff in from
systemic design type approaches is one
thing that's interesting is if you then
say humans like humans what we do is
as a child and maybe some apes do this
but not many is we point at something as
a child
and then we
point to the parent and that's becomes
it's in the niche but it becomes what
they would call a boundary object in
systemic design so you're trying to
bring together a team around something
which is beyond one discipline
so what you need to do is you say okay
we can't just use the
variational free energy landscape of
this
domain ecosystem which we have all
attuned to in our sort of own way
because now people are running by
different rules right we're running by
the physicists and the biologists right
they're just going to be so you get a
boundary object which you point to so
it's something in the niche
and then you sort of work to that
um so anyway i just think there's
something quite interesting then about
like daniel was saying there about
you know what sort of language and do we
use but
this scale problem comes back every time

but i i wondered um
like the difference between
a boundary object or something which can
be defined which i'm not sure if other
animals can really do
in the same way maybe a dog when it
urinates up a tree i was just thinking
about that does that become a boundary
object in the in their world i don't
know but um
yeah anyways something that comes to
mind
thanks steven and with the
transdisciplinarity
in cancer biology they've recognized the
niche
the tumor and the stem cell niche and
the micro
environment and there's so much more to
say on these very interesting topics
just in the last
few minutes we can go to a few of these
final um
points in the dot one it's always
awesome to have the author and unpack
some of the ideas
and so this is kind of our little
fractal.2
setting us up for our discussion next
week and thinking about implications and
how does it influence how we act moving
forward and what kinds of questions
we're looking at
you wrote about some of the implications
for thinking um
about
individuality from a cognitive
perspective

you say it broadly
and
it'd be awesome to hear your thoughts on
uh
the folks psychological notion of the
cognizer
and where does the fep come into play
right
um
thanks for the question
uh so the folk psychological notion of
cognizer
again
um my argument is a piggybacks
upon this idea of the
psychological
biological notion of the individual
and once we kind of free ourselves from
thinking about
uh individuals specifically in terms of
replication dynamics
uh that opens up a space
for us to think about
uh
cognizers
that are discontinuous with uh
replication or replicators
and i think this is like that that's
going to be very important
uh again
when taking the uh this particular route
of thinking about the evolution of
cognition
uh when thinking about for example okay
uh
if cognition goes all the way down
to a certain to a certain degree
uh

then what part does it actually play in
the the construction of larger systems
uh and why would why would the
construction of larger systems actually
be some type of adaptation
what uh for each of the systems involved
so

when we're thinking about for example i
think one particular thing that is quite
important will be quite important to
think about this for is
okay the manner in which we
think about artificial intelligence for
example

um
one way of course two ways of course of
doing this

uh is you know we go for various kind of
specific tasks

whether it's going to be kind of
computer vision or

uh gpt3 type tasks where we can actually
analyze large corpus of text

and kind of recapitulate that

uh that's one thing the other thing of

course is the task general or

kind of domain general

notion of ai

and i think in order to perhaps

tread into the department or turn into

that field of more kind of

task generality we're going to have to

think about even simpler questions as to

how can a system actually start

maintaining itself

uh according to its own tasks and again

we come back to this notion of how can

we actually

uh set a system up

you know to have its own intrinsic
motivation

uh

and one way to do this is perhaps to
piggyback on living systems so you
introduce a living system to an
artificial system

uh

and have to have them communicate in
such a way

uh that's

you know analogous to some type of
symbiotic system this is something again
perhaps that mike levin michael levin
has been thinking about doing actually
in his lab

uh and uh

and i think this this is quite an
interesting kind of method to think
about the importance of

uh of symbiosis in developing
artificially intelligent systems you
know um a system that can identify go go
to go to a room an embodied intelligent
agent that can like you know
traverse a room and identify all the
dogs that's great with it with an
accuracy a high accuracy level that's
fantastic but if

if they can't respond to the dog knowing
that a dog is going to actually you know
dismantle it i'm not sure if i would
want to call how intelligent that system
would even be in terms of having basic
intelligence

and i think that fep and active
influence allows allows a particular

type of
perspective on the notion of basic
intelligence or intelligence that's
surrounded or
uh
yes that's that's
grounded in the notion of kind of uh a
nest a nest density
or grounded in the notion of a type of
a moving set point that's equivalent to
a homeostatic
moving homeostasis moving from static
states and autostasis
uh supporting states
thank you matt shannon
thanks for that i wonder um
how important
sort of drawing on what flu said and
dave and steven i think also earlier
how important
some sort of effective or emotional
state or awareness is
in knowing and cognizing and um
like how important that is for for a
symbiotic relationship to be called a
cognizer
and whether you know like the neural
network that is working with the human
maybe to compose a piece of music um
how important it is for that neural
network to have
cognizing capabilities of its own or
maybe some sort of effective
awareness or some sort of effective
processing
that it is able to develop in some way
on its own or maybe only in symbiosis
with a person

um

and i don't know if if this bobtail
squid and the vibrio like do they have
sort of effective awareness together or
does this would seem to you know
experience something like
pain i suppose would be one of the most
basic effective states we might have
so how important would basically is
effective processing in
fi delineating individuals or cognizers
a great question shannon

um

i'll take a go at it um i do think that

uh

at a certain level of complexity
affective uh affective states play a
huge role

but i'm but because i'm i'm

i suppose i'm making an assumption that
cognition goes all the way down
and not every system is going to be an
affect have that capacity to have some
type of effect actual awareness
one might actually think about some
other type of uh the sub-component
that

plays a similar role as affect and
perhaps that just might be something
like um

you know

something that grounds

uh

that grounds a particular type of
motivation to stay uh to stay in
homostatic states or to remain in
particular homeostatic states
and this can be um

oh pamelia lyon has written about this
just recently actually um
and
it's not affect necessarily but it's
valence negative and positive valence
that might not and it has less to do
with awareness
or sentience in that sense uh but more
to do with actually
uh
a a particular type of organism
or particular type of system responding
in ways uh that
again
are responding in ways that keep it
centered
and whether or not this whether or not
that kind of validates
a bacteria might climb a gradient of
glucose for example
but bacterium might not have any type of
uh you know experience of that glucose
gradient
but somehow we might say that that
bacterium is actually it's it's
positively valenced it's
something that it's attracted to
right uh and maybe in maybe in simpler
systems you have something more like the
notion of valence rather than affect but
a certain period of uh you know
complexity after a certain perhaps a
degree of complexity arises
uh
affect is very important and i think
particularly when it comes into
multicellular systems
that

perhaps uh
that have to actually
compute
larger like
much more information has to compete
much more information and has to
interact in their environments their
environmental niches are much more uh
using a term by chem sterling much more
con opaque
so uh
effective information effective the
matter in which we can use affect and
how that effect how that affects
uh cognition
i think it becomes quite important
important but i'm not exactly sure
what what the
number one is probably not a line
uh it probably develops
but just what that function is is a
different story i don't know do you have
any ideas about that
i
don't have an idea about where that
would start but i feel like it would
fall off so we might have the small
scale like our bobtail squid and vibrio
who might have something like valence
that they're responding to and then as
we get slightly larger organisms you
know dogs people
then affect will play
a lot more of a role but then as we
have bigger systems like multiple humans
what is the effective like
state of a crowd of people like we can
ascribe like they look angry or they

sound angry but they as a crowd are they
feeling
something as a crowd kind of like group
intentionality is there a group
effective state it might kind of start
falling off again so you might have this
sort of u-shaped curve of where
valence matters and then somewhere here
in the sweet spot
effective states or emotions really
matter
that's a that's super
definitely something to absolutely
something i want to think about
oh and also reminiscent of the
discussion of affect as well as higher
order parameters in deep parametric
active inference where we even talked
about which states were opaque or not
whereas opaque usually meaning like you
can't see it you can't see in but
actually the opaque mental states are
the ones that we do experience so that
was in the sandved smith paper and
whether
other systems feel things what is it
like to be a squid what is it like to be
a symbiosis that's the phenomenology and
the metaphysics but
what is it like for us to model that
system as if there's a higher level
control parameter
we can know what that's like it's a lot
like this conversation
so just
everybody can provide a last
comment on this awesome discussion i
think the takeaway for me

is that in a um
post-darwin biology
it's easy to use the simplified trifecta
of evolution by natural selection
phenotypic variability heritability of
the phenotype fitness correlation and
variation in fitness and then
project that onto all the kinds of
individuality cognitive rights and
responsibilities the economic unit the
sentient unit all of those as just
simply under the coattails of the
darwinian replicator and i think
pluralism
and a bevy of other perspectives
suggests that that's just not the case
blue and then anyone else with a last
comment
i just wanted to say thanks um for
coming i really enjoyed this discussion
and i just started reading your paper
your 2021 paper with pizzullo uh the
modeling ourselves um and i hope you'd
consider coming back to talk about it
with us
thanks so much
it's been a really incredible
conversation discussion so
thank you all for inviting me
um shannon or stephen or dave any last
comments
awesome well
all participants thank you so much
and matt especially for writing the
paper and coming to talk about it and
seeing the dot zero before so
peace out everybody see you next week
bye

[Music]